Calculus: Simplified from first principles

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1 Critical Points and Derivative Tests

1.1 Critical Points

A critical point of a function $f(x)$ occurs where the derivative is either zero or undefined. These points are important because they often correspond to local maxima, local minima, or saddle points of the function.

Definition 1.1. *A point $x = c$ is called a **critical point** of a function $f(x)$ if either:*

$$f'(c) = 0 \quad \textit{or} \quad f'(c) \textit{ is undefined.}$$

To find critical points:

- Compute the derivative $f'(x)$.
- Set $f'(x) = 0$ and solve for x.
- Check where $f'(x)$ does not exist.

1.2 First Derivative Test

The **First Derivative Test** helps determine whether a critical point is a local maximum, a local minimum, or neither.

Theorem 1.1 (First Derivative Test)**.** *Let $f(x)$ be a continuous function and c a critical point.*

- *If $f'(x)$ changes from positive to negative at c, then $f(c)$ is a local maximum.*
- *If $f'(x)$ changes from negative to positive at c, then $f(c)$ is a local minimum.*
- *If $f'(x)$ does not change sign at c, then $f(c)$ is neither a local maximum nor a local minimum.*

Example 1.1. *Consider the function $f(x) = x^3 - 3x^2 + 2$. Find the critical points and classify them using the First Derivative Test.*

Solution:

$$\begin{aligned} f'(x) &= 3x^2 - 6x \\ &= 3x(x-2) \end{aligned}$$

Setting $f'(x) = 0$, we get critical points at $x = 0$ and $x = 2$.

Now, we examine the sign of $f'(x)$ around each critical point:

- *For $x = 0$: Choose $x = -1$ and $x = 1$.*

$$f'(-1) = 3(-1)(-1-2) = 9 \quad (\textit{positive})$$

$$f'(1) = 3(1)(1-2) = -3 \quad (\textit{negative})$$

So $f'(x)$ changes from positive to negative at $x = 0$, making it a local maximum.

- *For $x = 2$: Choose $x = 1.5$ and $x = 3$.*

$$f'(1.5) = 3(1.5)(1.5-2) = -2.25 \quad (\textit{negative})$$

$$f'(3) = 3(3)(3-2) = 9 \quad (\textit{positive})$$

So $f'(x)$ changes from negative to positive at $x = 2$, making it a local minimum.

1.3 Second Derivative Test

The **Second Derivative Test** provides another method to classify critical points by analyzing the concavity of the function.

Theorem 1.2 (Second Derivative Test)**.** *Let $f(x)$ be a function such that $f'(c) = 0$ and the second derivative $f''(x)$ exists.*

- *If $f''(c) > 0$, then $f(c)$ is a local minimum.*
- *If $f''(c) < 0$, then $f(c)$ is a local maximum.*
- *If $f''(c) = 0$, the test is inconclusive.*

Example 1.2. *Consider the function $f(x) = x^4 - 4x^3 + 4x^2$. Find the critical points and classify them using the Second Derivative Test.*

Solution:

$$\begin{aligned} f'(x) &= 4x^3 - 12x^2 + 8x \\ &= 4x(x^2 - 3x + 2) \\ &= 4x(x-1)(x-2) \end{aligned}$$

Setting $f'(x) = 0$, we get critical points at $x = 0$, $x = 1$, and $x = 2$.

Next, we compute the second derivative:

$$f''(x) = 12x^2 - 24x + 8$$

Evaluating at the critical points:

- *For $x = 0$, $f''(0) = 8 > 0$, so $x = 0$ is a local minimum.*
- *For $x = 1$, $f''(1) = -4 < 0$, so $x = 1$ is a local maximum.*
- *For $x = 2$, $f''(2) = 8 > 0$, so $x = 2$ is a local minimum.*

1.4 Practice Problems:

1. Find the critical points of $f(x) = x^3 - 3x^2 + 4x - 2$ and classify them using the First Derivative Test.
2. Determine the critical points of $f(x) = 2x^4 - 4x^3 + 2$ and use the Second Derivative Test to classify them.
3. For $f(x) = \sin(x) - \cos(x)$ on the interval $[0, 2\pi]$, find the critical points and classify them.
4. Compute the critical points of $f(x) = x^{2/3}(x-1)$ and determine their nature using the First Derivative Test.
5. Find the local maxima and minima of $f(x) = e^x - 2x$ using the First Derivative Test.
6. Determine and classify the critical points of $f(x) = \ln(x^2 + 1)$.
7. For $f(x) = \frac{x}{x^2+1}$, find the critical points and use the Second Derivative Test to classify them.
8. Determine the critical points of $f(x) = x^4 - 8x^3 + 18x^2$ and classify them using the First Derivative Test.
9. Find the critical points of $f(x) = \tan(x)$ on $[0, \pi]$, and classify them.
10. Compute the critical points of $f(x) = x^5 - 5x^3$ and classify them using the Second Derivative Test.
11. Find the critical points of $f(x) = \frac{x^2}{x^2-1}$ and classify them using the First Derivative Test.
12. Determine the critical points of $f(x) = \arctan(x) - \frac{x}{x^2+1}$ and classify them.
13. Compute the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 1$ and classify them using the First Derivative Test.
14. Find the local maxima and minima of $f(x) = x^2 e^x$ using the Second Derivative Test.
15. For $f(x) = \cos(x) + x^2$ on $[0, \pi]$, find the critical points and classify them.
16. Determine the critical points of $f(x) = x^3 - 3x^2 + x + 5$ and classify them using the First Derivative Test.
17. Find the critical points of $f(x) = \ln(x) - x$ and classify them using the Second Derivative Test.
18. Compute the critical points of $f(x) = \frac{x^2-1}{x}$ and determine their nature.
19. For $f(x) = e^{x^2-4x}$, find the critical points and classify them using the First Derivative Test.
20. Determine the critical points of $f(x) = \frac{1}{x} + \ln(x)$ and use the Second Derivative Test to classify them.
21. Let f be a function whose derivative is given by $f'(x) = (x-1)(x+3)(x-4)$. Where does the function f have local maxima and local minima?
22. What is the minimum value of the function $f(x) = \frac{3}{8}x^4 + x^3$?
23. Find the absolute maximum value of $f(x) = \frac{x+1}{x^2+3} on the interval [-2, 3]$.
24. Find the absolute minimum and the absolute maximum of the function $f(x) = \frac{lnx}{x^2}$ on the interval $[\frac{1}{e}, e]$.
25. A certain function $f(x)$ has the following first and second derivatives: $f'(x) = 3x^5 - 6x^3$, $f''(x) = 15x^4 - 18x^2$. How many relative maxima does $f(x)$ have?
26. Find the absolute maximum and minimum values of the function $f(x) = x^{\frac{8}{3}}$ on the interval $[-1, 8]$. If it does not exist, then state that it does not exist.
27. Find the absolute maximum and minimum values, on the interval $[0, 1]$ of the function $f(x) = cos(\pi x) + sin(\pi x)$